

Identification of Potential Adverse Drug Reactions using Random Walk on Network Models

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Abstract—Adverse drug event or adverse drug reactions (ADRs) are known to result from off-target pharmacological interactions and higher order effects. ADRs have been often 'discovered' only after clinical events are reported and archived in databases such as FAERS. Off-target interactions and higher order effects are mostly reported at the pairwise and second order levels, respectively. Hence, many likely-ADRs are overlooked for a long time, leading to a multitude of potential morbi-mortality events with enormous public and individual health consequences. With the advent of high throughput omic technologies and large public databases, an enormous corpus of biochemical and phenotypic data is available to study ADRs. Detailed analyses of such data corpora may allow us to develop a deeper understanding of these relevant, higher order interactions, leading to a better assessment of potential ADRs. However, extracting useful information from such large data conglomerates is a challenging endeavor for the data sciences. Here, we applied methods from knowledge discovery in databases, network pharmacology, supervised and semi-supervised statistical learning, to tackle the problem of identifying potential adverse drug reactions. By analyzing a set of 500 drugs, including 67 systemic anti-infective agents, we identified over 27,000 potential adverse drug reactions based on activity on well-known biological function sets.

Index Terms—Adverse drug reactions, network models, random walks on networks

I. INTRODUCTION

Adverse drug reactions are broadly defined as the set of undesired effects, ranging from simply unpleasant to potentially life threatening, associated to pharmacological treatment [1]. The origin of such drug reactions is multifactorial, with some being easily explained as exacerbated manifestations of the drugs desired effect, while others are not easily linked to the canonical targets of a given drug [2]. Off-target drug effects may potentially explain such adverse drug reactions, as well as help identify potential reactions not previously reported [3].

Extracting informative patterns from large data corpora is one of the goals of data mining or knowledge discovery in databases (KDD). Most KDD techniques were developed with tabular format databases, either relational or compositional, in mind. Further structured data sources allow for the use

of more refined knowledge discovery methods, often leading to deeper insights [4]. Theoretical approaches making use of probabilistic graphical models have been extensively used in the physical, biological and data sciences [5]. Once a knowledge graph has been built or inferred, graph theoretical and probabilistic methods can be used to further extract patterns from these data sources [6]–[8].

In this work we used a previously reported [3], [9] network model that integrates drugs, functional effects and adverse drug reactions. We used random walk ensembles [10] to identify and quantify the potential paths through which a given drug can generate an adverse drug reaction [11]. As a use case, we focus on anti-infectives for systemic use (ATC class J), as a class of drugs for which off-target effects are importantly associated to adverse effects [12]; we show that our network-based method identifies several potential off-label adverse drug effects [13]. Aside from off-target, higher order effects caused by off-target signals propagating via a series of molecular reactions/interactions, are a potential source of adverse events.

II. MATERIALS

Drug - Gene Ontology - Adverse drug reaction tripartite network: In this work we will use, as a starting point, a previously published [3], [9] network that was built by relying on probabilistic techniques (using principal angle enrichment analysis [14]) applied over very large datasets to link drug effects at the functional level (defined as Gene Ontology terms) to potential adverse drug reactions. This network has 3454 drugs, 323 Gene Ontology terms, and 111 Adverse Drug Effects.

The rationale behind this work is that a given drug is able to perturb a set of target and off-target biomolecules. The perturbed genes, proteins or metabolites in turn may be able to deregulate a set of *desired* (e.g. the therapeutic functions of the drug) and *undesired* (likely associated with the ADRs) processes as reported in the Gene Ontology (GO) database [15].

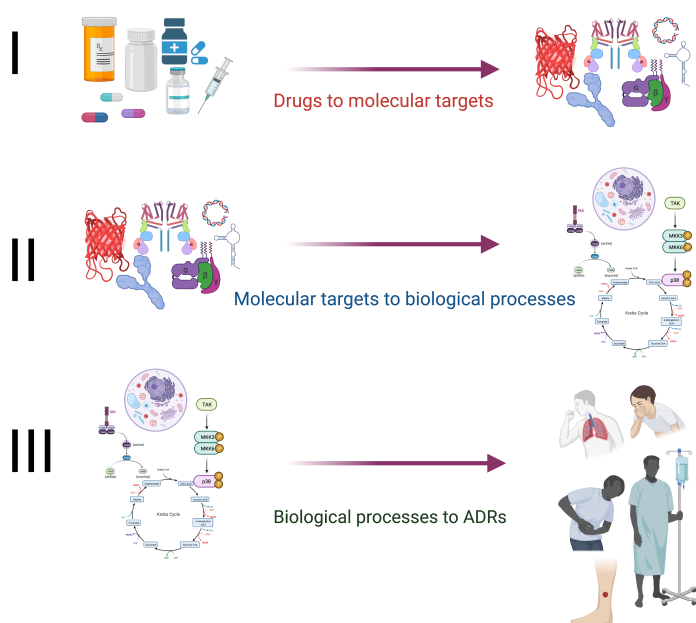


Fig. 1. The three dimensions of the tripartite network used in this work: **I** - Drugs are able to modify the state of the on- and off- molecular targets. **II** - Perturbed molecular targets change the state of biological processes. **III** - Altered biological processes may give rise to ADRs. Figure created with BioRender.com

In this sense, Gene Ontology entries provide a controlled vocabulary useful to describe several dimensions or biological features embedded in three main *branches*: Biological processes (BP), molecular function (MF) and cellular component (CC), grossly corresponding to the biological, physico-chemical and cell localization nature of such features [16]. Since we are interested in potential biological processes perturbed by the direct or indirect action of the drugs, the network is restricted to the BP branch of the Gene Ontology [17].

ADRs have, in turn, be related with the deregulation of certain biological processes [18], often by elucidating the underlying mechanisms of damage [19] or by modeling their biological network neighborhoods [20].

With this information, it is then likely to develop a networked approach to learning the relationships between pharmacological interventions and adverse reactions as follows: We can build directed and tripartite network: only **drug** $\rightarrow$ **GO(BP)** and **GO(BP)** $\rightarrow$ **ADR** links are found in the network.

SIDER 4.1: The SIDER Side Effect Resource [21] is a database that lists *on-label* drug side effects; that is, adverse drug reactions that are listed on labels and package inserts of commercially available products. On-label effects are the most immediate source of information for the prescribing

physician and the patient-consumer. Nevertheless, ADR reports that make it to the on-label regulatory stage do not comprehensively represent the full spectrum of ADRs. On the one hand, many ADR associations are yet to be discovered, documented, assessed and become regulated by organisms such as the FDA to finally make it to the on-label stage. On the other hand, only lower order effects are considered in the common regulatory enquiries, leaving out a plethora of potential, higher order effects.

Hence, in spite of its relevance in the clinical practice and its utility in the ascertainment of ADRs, SIDER entries cannot be considered to be *ground-truth* for computational discovery, but rather as a useful guideline.

III. METHODS

A. Random walks on networks

A relatively novel way to analyze networked data is by performing stochastic processes over the associated graph to unveil non-trivial information flow patterns in the connections encoded in such graph [22]. Random walks, in particular, have been extensively used due to their relative algorithmic simplicity and intuitive interpretability [23], [24]. Reconstructing the information flows from ensemble trajectories of random network is becoming a promising avenue for semi-supervised statistical learning [10], [25], [26] with potential applications in computational biomedicine [27]–[29].

It can be demonstrated that the structure of the tripartite, directed network limits the length of a random walk to two steps from an initial drug node, always ending on an ADR node. Therefore, each random walk is representing a potential ADR manifesting after the action of a given drug. With this in mind, our model is implemented as follows:

Algorithm 1 Drug to ADR random walk

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g ← tripartite drug GO ADR network
Drugs ← list of drugs
list.walks ← for d in Drugs do
  for i in 1:iters do
    walk = RandomWalk(start = d, length = 2)
    return last.node(walk)
  end for
end for
return list.walks

```

B. Identification of off-label reactions

While identifying on-label and off-label reactions are important problems, we are more interested in having a method that identifies off-label reactions, since while they are plausible, they are less likely to be considered in clinical settings.

To do so, we search each of the reactions identified with the random walk method in SIDER, a database that records on-label adverse drug reactions. With this information we

calculate, for each drug, the percentage of off-label ADRs identified out of the total predicted ADRs.

IV. RESULTS AND DISCUSSION

A. Profiles of Drug ADRs

We profiled 500 commercially available drugs reported in the original tripartite network. For the purposes of discussion, we illustrate our findings on anti-infective drugs (ATC class J), since by definition, the molecular targets of these drugs (within the microbial organisms) are not the same as the same as the functional targets through which most adverse drug reactions occur. As it is illustrated in Figure 2, there are many possible paths that can lead from a given drug to a given adverse reaction. In this work, to strike a balance between state-space coverage and analysis time, we performed 500 iterations per drug.

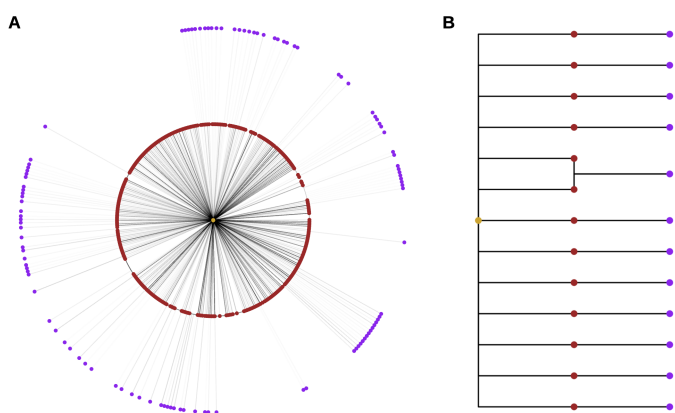


Fig. 2. Network visualization of the tripartite drug-Gene Ontology-adverse drug reaction network. In panel A, Drug nodes are colored in gold at the center of the network; gene ontology terms (associated with biological functions) are located in the middle circle, in red, and adverse drug reactions are located in the outer circle in purple. Panel B shows a subgraph for a single drug.

It is in this context that random walk parsing of data-rich graphs allow for a computationally feasible means to navigate the combinatorially very large space of drugs-molecules-processes-adverse reactions that, at the same time provides significance bounds to prioritize the more relevant hits [30].

In brief, through our random walk model, we have been able to identify which adverse drug reactions are more likely to be associated with each particular drug. The use of random walk trajectory frequencies as a means of prioritizing path-queries over a data-rich graph structure has been applied previously within the computational biomedicine setting, for instance as a means to score cancer genes in tumor-related networks [31], to predict diseases associated with perturbed miRNA profiles in OMIM disease similarity networks [32] and to assess disease genes in genotype-phenotype maps [33].

In Figure 3 we illustrate these in a *heatmap*, in which each row represents an (anti-infective) drug and each column

represents an adverse drug reaction; the color intensity (see color scale) of each square is proportional to the number of walks that reaches the ADR from the drug. Each row in this heatmap can be thought to represent the *ADR risk profile* of a given row. Intuitively, a drug is more likely to generate an ADR if it has a large number of off-target hits. By resorting to a stochastic walk over the network, we are able to assess whether a given drug is potentially able to reach a certain number of targets, and thus via the trajectories on the graph, being associated with a number of ADRs. The larger the territory that a given drug walk covers of the tripartite network search space, the larger its number of off-target hits, in turn, the larger the number of off-target interactions, the more likely is for an ADR being present.

Hence, by random walk parsing of our tripartite network, we are able to have a *quantitative* measure of the likelihood that a drug may lead to an ADR. This quantitative measure or score, corresponds precisely to the color intensity of the cells of the heatmap in figure 3. A close analysis of figure 3 also reveals an interesting pattern. Most of the ADRs analyzed have only a few associated drugs emerging from their random walk traversing of the tripartite network, as represented by the mostly-blue columns. However, a number of adverse effects present a much wider spectrum of associated drugs as represented by the yellowish columns. We can also notice that larger frequencies of ADRs as revealed by reddish and orange cells occur for these same, *systemic* effect drugs.

It is, precisely for these system-wide effect, larger ADR frequency drugs that the treating physician should be more cautious to prescribe, likely after a comprehensive assessment of the risk and benefits of treatment. Such kind of phenomena have been recently studied in the context of systems pharmacology [34]. In brief, systems pharmacology is focused on applying systems biology and network science to identify drug targets, to understand drug action in the context of cellular and molecular networks and to predict main mechanisms of action, off-targets and adverse drug effects.

In the case of the pharmacological and drug-development specialists, figure 3, in particular column clustering patterns, may provide useful information as to *families* of drugs (perhaps chemically or mechanistically-related) that are similar in their potential to unleash undesired reactions leading to ADRs. This may provide clues as to what to enhance and what to avoid while formulating new treatments or adjusting dosage. This same information may also be useful for pharmacological health policy officials as a means to develop differential regulatory mechanisms that will be guided by the likelihood of adverse effects [35].

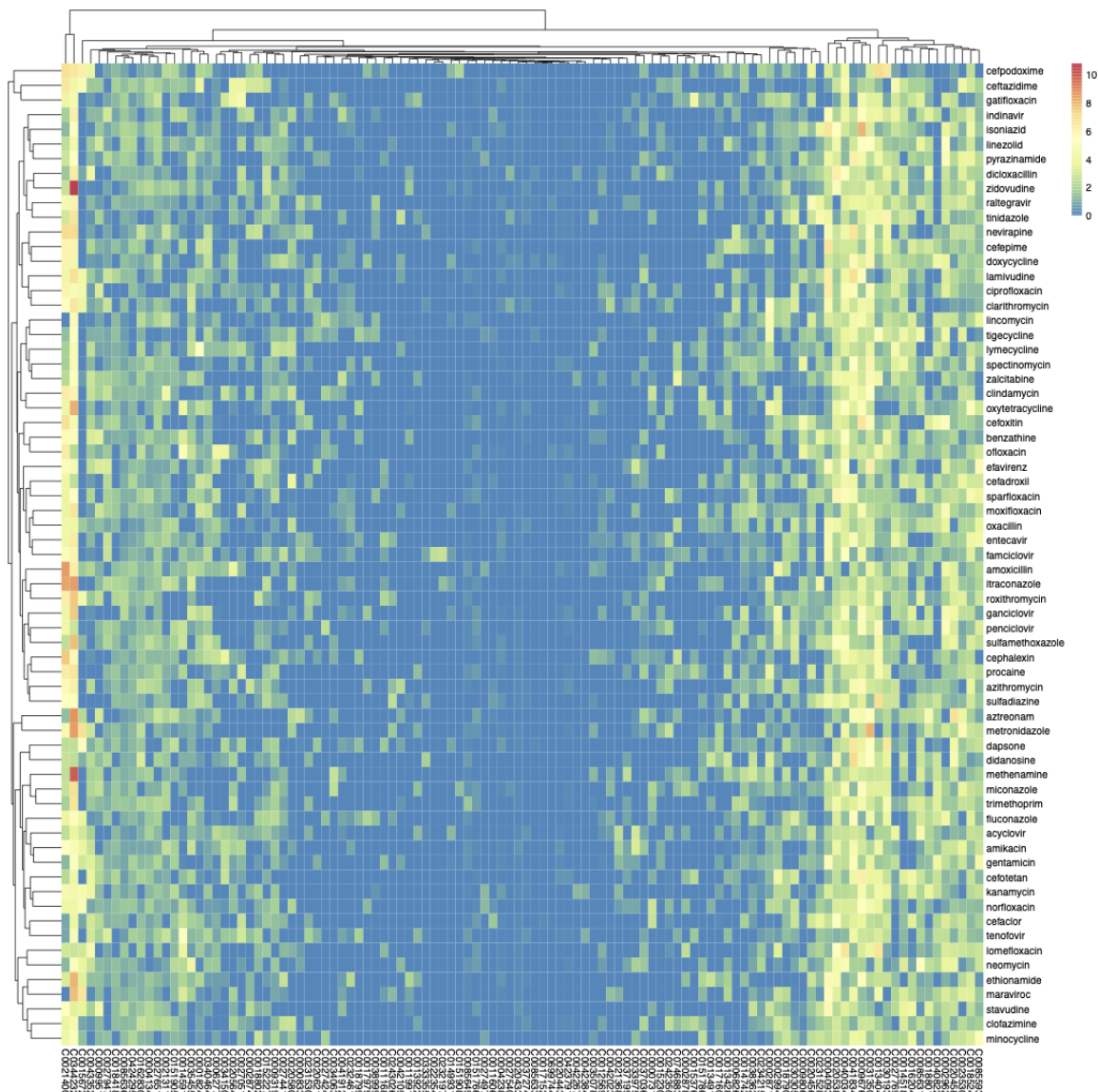


Fig. 3. Heatmap of anti-infective drug ADR risk profile.

B. Identification of off-label ADRs

We explored whether the identified potential ADRs are more likely to be those that have been previously reported and are included in the information on commercially available formulations (*on-label*) or whether our method is discovering new potentially important ADRs. It should be highlighted that we are not considering *on-label* effects to represent a *ground truth* on whether a drug is associated with a known side effect; inclusion into labels follows a complex pharmacovigilance and regulatory process, whereas our method is trying to capture the *plausibility* of a mechanistic link between drugs and adverse drug reactions.

In figure 4, we show the number of ADRs identified by our method for each (anti-infective) drug, and the percentage of said ADRs that is identified as being *off-label*. We observed

that our method is most frequently capturing *off-label* effects, that those marked in the existing databases, which may them susceptible of not being considered in a clinical setting. It is in this context that the major relevance of data-driven studies such as this one may provide a major contribution [36]–[38].

It is worth noticing that the biological and pharmacological databases from which the tripartite network was constructed are constantly being updated. Knowledge discovery in databases, including data-rich graph analyses such as the one presented here, are only as good and comprehensive as the input data is. Hence, it is expected that even without further methodological improvement, the present results may be enhanced in their comprehensiveness and trustability by keeping up with the periodic update of its database sources.

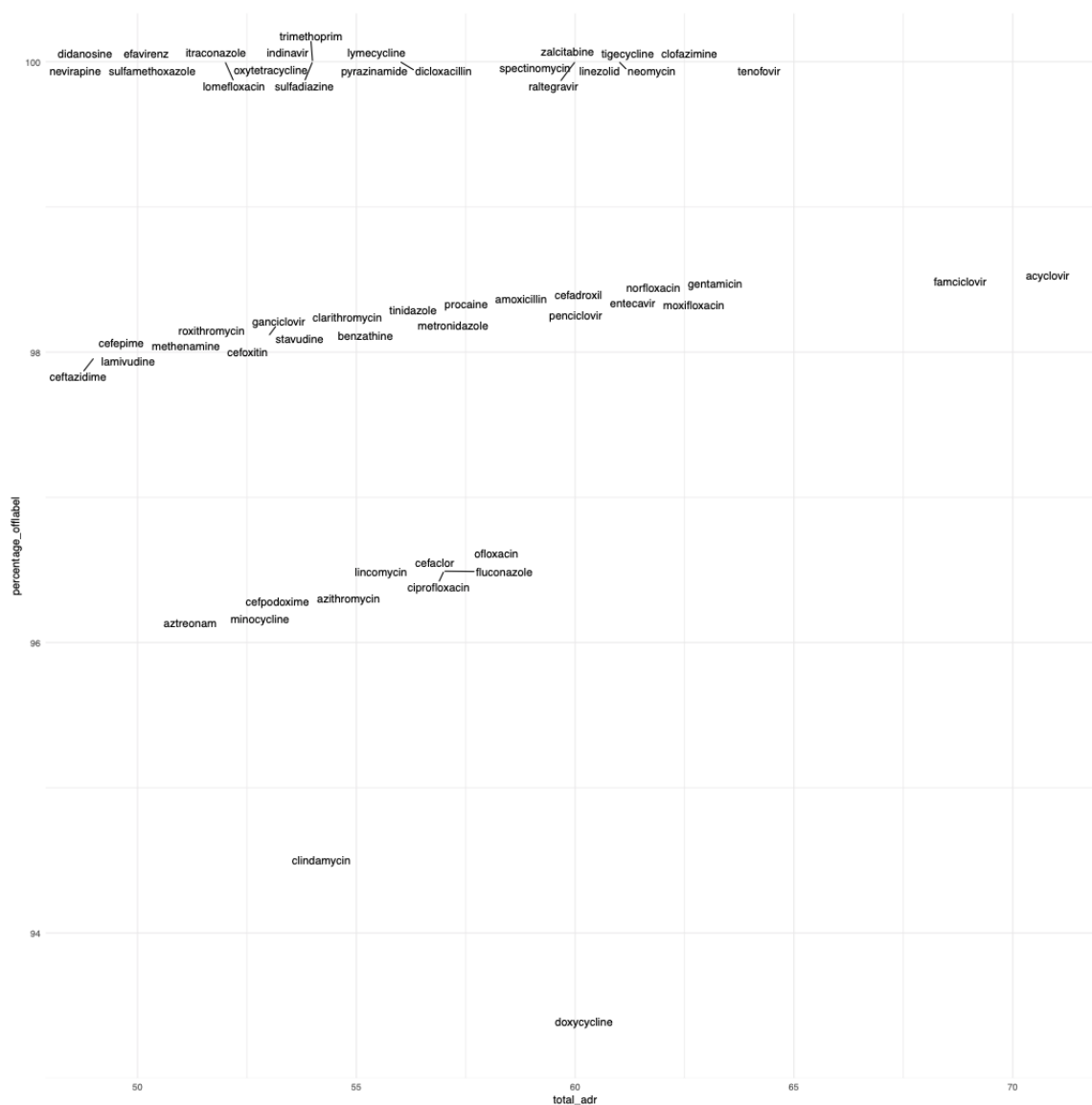


Fig. 4. Scatter plot of anti-infective drugs: number of total ADRs vs percentage of off-label ADRs.

V. CONCLUSIONS

In this work we have analyzed how stochastic processes over data-enriched graphs provide powerful means to extract information encoded in such graphical models. In particular, by generating random walk ensembles of trajectories over a tripartite **Drug** → **GO(BP)** → **ADR** directed network built from data on adverse reactions to class J ATC anti-infective drugs, we were able to find a much larger space of adverse drug reactions than those previously reported.

Aside from their theoretical relevance, the use of such statistical learning on graph methods in the context of adverse drug reactions is of large practical importance since adverse reactions to pharmacological treatment are a cause of concern to public health officials, health systems, pharmaceutical companies, but also, to the general public that may be

suffering from ADR related illnesses.

It is worth noticing that the network based approach just presented is based on an analysis of large data corpora that are constantly being updated (genetic, pharmacological, etc.), as such, these methods are contingent on the availability of high quality data sources. Any clinical use of the methods must consider the possibility of false positive associations and deal with statistical assessment of these. Network based studies however, are indeed more robust than traditional regression studies in view of the large redundancy of statistical dependencies in biological networks.

In the particular example that we have considered here. By studying a set of 500 drugs, of which 67 are systemic anti-infective agents, we were able to identify more than

27,000 potential adverse drug reactions based on activity on well-known biological function sets. Though these are to be considered *potential* ADRs as of this stage. Allowing the caring physicians and the patients to know of these potential effects in advance may contribute to personalized pharmacovigilance thus empowering the patient and diminishing the burden of ADR care for the health systems.

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CODE AVAILABILITY

Code for these analyses and links to the original datasets can be found at https://github.com/guillermodeandajauregui/adr_randomwalks

REFERENCES

- [1] J. Sultana, P. Cutroneo, and G. Trifirò, "Clinical and economic burden of adverse drug reactions," *Journal of pharmacology & pharmacotherapeutics*, vol. 4, no. Suppl1, p. S73, 2013.
- [2] J. Ritter, R. J. Flower, G. Henderson, Y. K. Loke, D. J. MacEwan, and H. P. Rang, *Rang and Dale's pharmacology*. Elsevier, 2019.
- [3] G. de Anda-Juregui, K. Guo, and J. Hur, "Network-based assessment of adverse drug reaction risk in polypharmacy using high-throughput screening data," *International Journal of Molecular Sciences*, vol. 20, no. 2, 2019.
- [4] R. Angles and C. Gutierrez, "Survey of graph database models," *ACM Computing Surveys (CSUR)*, vol. 40, no. 1, pp. 1–39, 2008.
- [5] E. Hernández-Lemus, "Random fields in physics, biology and data science," *Frontiers in Physics*, vol. 9, p. 77, 2021.
- [6] R. Angles, M. Arenas, P. Barceló, A. Hogan, J. Reutter, and D. Vrgoč, "Foundations of modern query languages for graph databases," *ACM Computing Surveys (CSUR)*, vol. 50, no. 5, pp. 1–40, 2017.
- [7] Y. Liang and P. Zhao, "Similarity search in graph databases: A multi-layered indexing approach," in *2017 IEEE 33rd International Conference on Data Engineering (ICDE)*, pp. 783–794, IEEE, 2017.
- [8] N. Francis, A. Green, P. Guagliardo, L. Libkin, T. Lindaaker, V. Marsault, S. Plantikow, M. Rydberg, P. Selmer, and A. Taylor, "Cypher: An evolving query language for property graphs," in *Proceedings of the 2018 International Conference on Management of Data*, pp. 1433–1445, 2018.
- [9] Z. Wang, N. R. Clark, and A. Maayan, "Drug-induced adverse events prediction with the lincs l1000 data," *Bioinformatics*, vol. 32, no. 15, pp. 2338–2345, 2016.
- [10] H. Jiang, Y. Song, C. Wang, M. Zhang, and Y. Sun, "Semi-supervised learning over heterogeneous information networks by ensemble of meta-graph guided random walks," in *IJCAI*, pp. 1944–1950, 2017.
- [11] R. Xue, J. Liao, X. Shao, K. Han, J. Long, L. Shao, N. Ai, and X. Fan, "Prediction of adverse drug reactions by combining biomedical tripartite network and graph representation model," *Chemical research in toxicology*, vol. 33, no. 1, pp. 202–210, 2019.
- [12] L. Aagaard, L. H. Nielsen, and E. H. Hansen, "Consumer reporting of adverse drug reactions," *Drug safety*, vol. 32, no. 11, pp. 1067–1074, 2009.
- [13] T. Eguale, D. L. Buckeridge, A. Verma, N. E. Winslade, A. Benedetti, J. A. Hanley, and R. Tamblyn, "Association of off-label drug use and adverse drug events in an adult population," *JAMA internal medicine*, vol. 176, no. 1, pp. 55–63, 2016.
- [14] S. Van Dijk, P. Molloy, H. Varinli, J. Morrison, and B. Muhlhausler, "Epigenetics and human obesity," *International journal of obesity*, vol. 39, no. 1, pp. 85–97, 2015.
- [15] G. O. Consortium, "Expansion of the gene ontology knowledgebase and resources," *Nucleic acids research*, vol. 45, no. D1, pp. D331–D338, 2017.
- [16] G. O. Consortium, "The gene ontology resource: 20 years and still going strong," *Nucleic acids research*, vol. 47, no. D1, pp. D330–D338, 2019.
- [17] P. D. Thomas, "The gene ontology and the meaning of biological function," in *The Gene Ontology Handbook*, pp. 15–24, Humana Press, New York, NY, 2017.
- [18] S. M. Ivanov, A. A. Lagunin, and V. V. Poroikov, "In silico assessment of adverse drug reactions and associated mechanisms," *Drug Discovery Today*, vol. 21, no. 1, pp. 58–71, 2016.
- [19] M. R. Boland, A. Jacunski, T. Lorberbaum, J. D. Romano, R. Moskovitch, and N. P. Tatonetti, "Systems biology approaches for identifying adverse drug reactions and elucidating their underlying biological mechanisms," *Wiley Interdisciplinary Reviews: Systems Biology and Medicine*, vol. 8, no. 2, pp. 104–122, 2016.
- [20] H. Rahmani, G. Weiss, O. Méndez-Lucio, and A. Bender, "Arwar: A network approach for predicting adverse drug reactions," *Computers in biology and medicine*, vol. 68, pp. 101–108, 2016.
- [21] M. Kuhn, I. Letunic, L. J. Jensen, and P. Bork, "The sider database of drugs and side effects," *Nucleic acids research*, vol. 44, no. D1, p. D1075D1079, 2016.
- [22] R. van der Hofstad, "Stochastic processes on random graphs," *Lecture notes for the 47th Summer School in Probability Saint-Flour*, 2017.
- [23] N. Masuda, M. A. Porter, and R. Lambiotte, "Random walks and diffusion on networks," *Physics Reports*, vol. 716, pp. 1–58, 2017.
- [24] X. Huang, Q. Song, Y. Li, and X. Hu, "Graph recurrent networks with attributed random walks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 732–740, 2019.
- [25] P. Vernaza and M. Chandraker, "Learning random-walk label propagation for weakly-supervised semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7158–7166, 2017.
- [26] N. Lao, T. Mitchell, and W. Cohen, "Random walk inference and learning in a large scale knowledge base," in *Proceedings of the 2011 conference on empirical methods in natural language processing*, pp. 529–539, 2011.
- [27] G. Yu, G. Fu, C. Lu, Y. Ren, and J. Wang, "Brwlda: bi-random walks for predicting lncrna-disease associations," *Oncotarget*, vol. 8, no. 36, p. 60429, 2017.
- [28] S. Y. Kim, H.-H. Jeong, J. Kim, J.-H. Moon, and K.-A. Sohn, "Robust pathway-based multi-omics data integration using directed random walks for survival prediction in multiple cancer studies," *Biology direct*, vol. 14, no. 1, pp. 1–13, 2019.
- [29] X. Chen, M.-X. Liu, and G.-Y. Yan, "Drug–target interaction prediction by random walk on the heterogeneous network," *Molecular BioSystems*, vol. 8, no. 7, pp. 1970–1978, 2012.
- [30] M. Gardner, P. Talukdar, J. Krishnamurthy, and T. Mitchell, "Incorporating vector space similarity in random walk inference over knowledge bases," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pp. 397–406, 2014.
- [31] P.-J. Wei, F.-X. Wu, J. Xia, Y. Su, J. Wang, and C.-H. Zheng, "Prioritizing cancer genes based on an improved random walk method," *Frontiers in genetics*, vol. 11, p. 377, 2020.
- [32] H. Chen and Z. Zhang, "Prediction of associations between omim diseases and micrnas by random walk on omim disease similarity network," *The Scientific World Journal*, vol. 2013, 2013.
- [33] Y. Li and J. Li, "Disease gene identification by random walk on multigraphs merging heterogeneous genomic and phenotype data," in *BMC genomics*, vol. 13, pp. 1–12, BioMed Central, 2012.
- [34] E. K. Silverman and J. Loscalzo, "14. systems pharmacology in network medicine," in *Network Medicine*, pp. 324–340, Harvard University Press, 2017.
- [35] R. M. Gallagher, J. J. Kirkham, J. R. Mason, K. A. Bird, P. R. Williamson, A. J. Nunn, M. A. Turner, R. L. Smyth, and M. Pirmohamed, "Development and inter-rater reliability of the liverpool adverse drug reaction causality assessment tool," *PloS one*, vol. 6, no. 12, p. e28096, 2011.
- [36] T.-B. Ho, L. Le, D. T. Thai, and S. Taewijit, "Data-driven approach to detect and predict adverse drug reactions," *Current pharmaceutical design*, vol. 22, no. 23, pp. 3498–3526, 2016.
- [37] R. Liu, M. D. M. AbdulHameed, K. Kumar, X. Yu, A. Wallqvist, and J. Reifman, "Data-driven prediction of adverse drug reactions induced by drug–drug interactions," *BMC Pharmacology and Toxicology*, vol. 18, no. 1, pp. 1–18, 2017.
- [38] N. P. Tatonetti, P. Y. Patrick, R. Daneshjou, and R. B. Altman, "Data-driven prediction of drug effects and interactions," *Science translational medicine*, vol. 4, no. 125, pp. 125ra31–125ra31, 2012.